Rasul Khanbayov

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Education

Hamad Bin Khalifa University — Doha, Qatar
Ph.D. in Computer Science & Engineering — Jan. 2026 – Present

- Research: physical reasoning from video and reliability of vision–language models

Technical University of Munich — Munich, Germany
M.Sc. in Informatics — Oct. 2022 – Aug. 2025

- Major: Machine Learning & Analytics. Minors: Computer Graphics & Vision; Engineering Software-intensive Systems
- Thesis: *Multimodal Registration of 3D OCT and Fundus Images of the Retina*

Eötvös Loránd University — Budapest, Hungary
B.Sc. in Computer Science — Sep. 2019 – Jun. 2022

- GPA: 5.0/5.0 (highest grade)
- Thesis: *Orchestrator Cloud Manager Application*

Research Interest

Physical Reasoning from Video, Vision–Language Models, Multimodal Learning, Reliable Evaluation and Benchmarks, 3D/4D Scene Understanding, Medical Imaging

Preprints & Publications

*denotes equal contribution

Selected Publications

1. **Rasul Khanbayov**, Erchin Serpedin, Hasan Kurban. GRAPE: Graph-Augmented Prototype Explanations for Interactive Medical Image Diagnosis. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
2. **Rasul Khanbayov***, Mohamed Rayan Barhdadi*, Erchin Serpedin, Hasan Kurban. IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video. *European Conference on Computer Vision (ECCV)*, Springer, pp. 366–383, 2026.
3. **Rasul Khanbayov**, Hasan Kurban. Consistency Has a Computable Blind Spot: A Commutation Theory of Label-Free Reliability for Vision–Language Figure Reading. *Findings of the Association for Computational Linguistics: EACL*, 2027.
4. **Rasul Khanbayov**, Hasan Kurban. When Does Consensus Mean Correctness? Measuring the Agreement–Accuracy Coupling with Semantics-Preserving Re-Rendering. *Findings of the Association for Computational Linguistics: EACL*, 2027.

Other Peer-reviewed Publications

5. **Rasul Khanbayov**, Hasan Kurban. Topical Phase Transitions in Artificial Intelligence Research: Large-Scale Evidence and an Early-Warning Signature for Emerging Topics. *Quantitative Science Studies*, 2026.
6. Hasan Kurban, **Rasul Khanbayov**, Mustafa Kurban. Physics as the label for measuring and correcting materials reasoning in multimodal models. *Knowledge-Based Systems*, 2026.
7. Danqing Chen*, Adithi Satish*, **Rasul Khanbayov***, Carolin M. Schuster, Georg Groh. Tuning Into Bias: A Computational Study of Gender Bias in Song Lyrics. *Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature (LaTeCH-CLfL)*, pp. 117–129, 2025.

8. Guohao Lin, Shidong Pan, **Rasul Khanbayov**, Changxi Yang, Ani Khaloian-Sarnaghi, Andriy Kovryga. Automated Knot Detection and Pairing for Wood Analysis in the Timber Industry. *International Conference on Neural Information Processing (ICONIP)*, Springer, pp. 81–95, 2025.

Under Review

9. **Rasul Khanbayov**, Hasan Kurban. Truncation Samplers Are Nested: Comparing Decoding Rules Is a Measurement Problem. *Submitted to ICLR 2027* (arXiv:2610.15570).
10. **Rasul Khanbayov**, Hasan Kurban. The Gradient That Top-k Distillation Never Computes. *Submitted to AISTATS 2027* (arXiv:2610.15589).
11. **Rasul Khanbayov**, Hasan Kurban. A Low Grounding Score Is Not an Ungrounded Judge: Identifying the Perceptibility Confound in Multimodal Oversight. *Submitted to AAAI 2027, AI Alignment Track* (arXiv:2610.00111).
12. **Rasul Khanbayov**, Mariam Sohail, Ahmed Abdala, Hasan Kurban. One Perturbation Is Not Enough: Identifiability and Blind Baselines for Behavioral AI Evaluation. *Submitted to AAAI 2027, AI Alignment Track* (arXiv:2609.06190).
13. **Rasul Khanbayov**, Hasan Kurban. PhysWeep: Does a Video Generator Realize the Physics You Ask For? *Submitted to Pattern Recognition* (arXiv:2609.06207).

Mentored Student Research

Papers led by undergraduate students in our lab whom I mentor as a PhD student; I contributed in an advisory role.

14. Rama AlHamidi, **Rasul Khanbayov**, Erchin Serpedin, Hasan Kurban. Counterfactual Sensitivity Is Not Repairability: Auditing Replay Probes for Video Evidence. *arXiv preprint arXiv:2608.15685*, 2026.
15. Aseel Mohamed, **Rasul Khanbayov**, Erchin Serpedin, Hasan Kurban. Conformal Coverage Guarantees for Any Video Temporal Grounder. *Submitted to AISTATS 2027* (arXiv:2608.07434).
16. Rama AlHamidi, Aseel Mohamed, **Rasul Khanbayov**, Mohamed Rayan Barhdadi, Erchin Serpedin, Hasan Kurban. Accuracy and Cost Claims Do Not Survive Re-Execution in Agentic VideoQA. *Submitted to Image and Vision Computing* (arXiv:2607.01469).
17. Mohamed Rayan Barhdadi, Samir Abdaljalil, **Rasul Khanbayov**, Erchin Serpedin, Hasan Kurban. 4D Synchronized Fields: Motion-Language Gaussian Splatting for Temporal Scene Understanding. *Submitted to AAAI 2027* (arXiv:2603.14301).

Research Experience

Graduate Research Assistant Jan. 2026 – Present
Hamad Bin Khalifa University, Doha, Qatar

- Research in computer vision and multimodal learning, focused on physical reasoning from video and on vision–language understanding of charts and scientific figures.
- Physics from video: recovering the dynamics and physical parameters of real-world systems from monocular video, including a benchmark for inverse system identification (ECCV 2026) and motion-language Gaussian splatting for temporal scene understanding.
- Charts and figures: label-free reliability measures for vision–language figure reading, testing when model agreement actually implies correctness via semantics-preserving re-rendering.
- Mentor undergraduate students in the lab on video agents, temporal grounding and 4D scene understanding.

Student Researcher Sep. 2023 – Aug. 2024

Technical University of Munich, Munich, Germany

- Computer vision for wood technology: automated knot detection and pairing on high-resolution board images; fine-tuned YOLOv8 (mAP@0.5 of 0.887) and paired knots via feature-embedding clustering (ICONIP 2025).
- NLP & computational social science: BERTopic pipeline over large-scale song lyrics and gender-bias measurement with Word2Vec and SC-WEAT (LaTeCH-CLfL 2025).

Industry Experience

Artificial Intelligence Engineer Nov. 2025 – Aug. 2026
OneVue Inc., Cleveland, USA (Remote)

- Built and optimised retinal image analysis pipelines for early detection of diabetic and pre-diabetic conditions.
- Adapted retinal foundation models (RETFound, ViT-based) through transfer learning and fine-tuning for downstream classification, reducing reliance on large labelled clinical datasets.
- Combined fundus/OCT imaging with patient metadata (multimodal inputs) to improve diagnostic performance.

Computer Vision Engineer Sep. 2024 – Aug. 2025
SynthesEyes, Munich, Germany

- Designed attention U-Nets mapping high-resolution 3D intraoperative OCT volumes to informative 2D feature maps for real-time surgical visualisation, outperforming conventional OCT projections.
- Built a multimodal registration pipeline with LoFTR and GeoFormer to align learned 2D OCT representations with complementary imaging views.

Business Analyst Intern Oct. 2022 – Mar. 2023
BMW Group, Munich, Germany

- Built KPI dashboards for product-line quality management, used by quality managers to surface and track issues.
- Debugged, tested and refactored the existing codebase (JavaScript, Python, PySpark, SQL, Slate).

Software Developer (Student) Mar. 2021 – Jul. 2022
Ericsson, Budapest, Hungary

- Developed and maintained features for the Ericsson Orchestrator Cloud Manager in a cross-functional team; wrote functional tests and resolved customer-raised trouble reports (Java, Spring, Docker, Kubernetes, Jenkins).

Awards and Honors

- **Qatar Foundation Scholar**, Ph.D. scholarship Jan. 2026
- **Azerbaijani State Program Scholarship**, M.Sc. scholarship Aug. 2022
- **Outstanding Student of the Faculty**, Eötvös Loránd University May 2022
- **Stipendium Hungaricum Scholar**, B.Sc. scholarship, Hungary Sep. 2020

Service

Reviewer: AAMAS, AAAI, ICLR 2027

Teaching

- **Tutor Assistant**, ENGR 110: Introduction to Programming
  Hamad Bin Khalifa University Spring 2026
- **Demonstrator**, IP-18fKVADA: Ada Programming Language
  Eötvös Loránd University Winter 2021, Spring 2022
- **Tutor Assistant**, IP-18fFUNPEG: Functional Programming Language
  Eötvös Loránd University Spring 2021

Skills

Programming Languages: Python, C++, Java, JavaScript, SQL
ML Frameworks: PyTorch, TensorFlow, Hugging Face, Ultralytics, ONNX, timm, MONAI, scikit-learn, NumPy, SciPy
Computer Vision: OpenCV, CUDA, detection/tracking, Vision Transformers, U-Net, image registration (LoFTR, GeoFormer), 3D/Gaussian splatting, medical imaging (OCT/fundus, DICOM/NIfTI)
NLP & LLMs: vision–language models, BERT/GPT, RAG, LlamaIndex, Ollama, BERTopic, prompt engineering
Tools: Git, Docker, Kubernetes, Jenkins CI/CD, AWS, Azure